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active learning in higher education

# The impact of formation and diversity on student team conflict

Active Learning in Higher Education
2024, Vol. 25(3) 409–423
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DOI: 10.1177/14697874221144998
journals.sagepub.com/home/alh
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## Abstract

Instructors of college level business courses utilize group assignments to stress the importance of collaborative work in professional organizations for students. How instructors determine team formation, whether assigning students to groups or allowing students to form their own groups, could impact the effectiveness of the group regarding their cohesiveness, conflict, or social loafing. Best practices suggest that instructors strategically assign students to teams to maximize the diversity of the members. However, could the diversity of group members contribute to conflict amongst members? Could the formation method contribute to the members' functioning as a team or group? The research questions in the study are: To what degree does team formation method impact (1) members' conflict; (2) members' cohesiveness; (3) members' social loafing; and (4) members' functioning as a team or group? Over 2 years, simultaneous sections of the same management course utilized different team formation procedures. Each year in one section, the instructor designed teams to maximize diversity of members. In the other course section, students freely formed their own teams. The findings suggest that instructor-designed diverse teams did not increase member conflict. Also, student-selected teams did not improve team cohesiveness. However, instructor-designed diverse teams did impact social loafing and the members' functioning as a team.

## Keywords

collaborative assignments, higher education, social loafing, team diversity, team formation

This study analyzed the impact of team formation and diversity on conflict and cohesiveness. Most colleges of business assign students group projects as part of their efforts to prepare students for the collaborative working world. Collaborative learning, the form of active learning that emphasizes student interaction, is regularly used in management courses (Prince, 2004).

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Collaborative learning offers many benefits including improved critical thinking skills and student engagement (Saiz Sanchez et al., 2014). Further, the content of organizational behavior management courses addresses the importance of group and team working within professional organizations. Utilizing teamwork within such a course reinforces course content to students in the class.

Two different theories of teamwork have hypothesized the relationship between team diversity and performance—social-psychology's similarity-attraction paradigm (Byrne, 1971) and management's cognitive resource diversity theory (Cox and Blake, 1991). The two theories have made opposite arguments in regard to optimizing team performance. The two theories have opposing positions in regard to diversity's impact on team performance.

The similarity-attraction theory has suggested that members' perceptions of others, typically in terms of demographics, leads to team cohesiveness (O'Reilly et al., 1989; Smith et al., 1994; Wiersema and Bantel, 1992). This theory has argued against diversity and has suggested that homogeneous teams are likely to be more productive because of team member mutual attraction, while heterogeneous groups would be less productive and have lower team cohesiveness because of relational conflicts arising from member differences (Horwitz, 2005).

Researchers of cognitive resource diversity theory have argued that diversity has a positive impact on team performance, as members bring unique cognitive resources to the team (Cox and Blake, 1991; Hambrick et al., 1996). Proponents of diversity have argued that teams with heterogeneous members have improved creativity, innovation, problem-solving, and decision-making. Organizations that have taken steps to avoid groupthink have been the most innovative (Janis, 1972). Further, organizations' diversity efforts have contributed positively to their financial bottom lines (Dodds et al., 2012.).

College professors striving to ready their students for employment have appreciated the value organizations place on hiring team players. Instructors in business colleges attempt to teach students to the benefits of teamwork by assigning group projects. Whether those students work collectively as a team with shared goals or as a group of individuals could impact the members' performance. Researchers have identified that the difference between groups and teams can be very different, and how the members understand this difference can be important (Armstead et al., 2016).

This study sought to analyze the impact of instructor-designed diverse student teams on team conflict, cohesiveness, social-loafing, and performance as a team or group of individuals. The study investigated whether instructor-designed diverse teams or student-selected teams differed in their perceived member conflict, cohesiveness, social loafing, and performance as a team. The study evaluated 2 years of concurrent courses with parallel courses each year with team formation procedures set per class.

## Literature review

The ability to be a team player is one of the top characteristics that employers desire in a prospective employee (Ashraf, 2004). College and university instructors strive to exemplify the importance of group and team work to students by assigning group projects within courses. While instructors may have the best intentions for their students, group work can be challenging for students as conflicts and perceived lack of effort by some members impacts the students' experience.

Instructors often note that when students are enabled to self-select project group members, a students tend to select friends of shared major, gender, and race (Oakley et al., 2004). However,

when teachers assign students to groups with diversity in their majors, genders, and races, students' group formation is based on an assignment rather than self-selection of friendly classmates. The goal of this study was to evaluate the impact of instructor-assigned diverse teams on members' perceived conflict, cohesiveness, social loafing, and functioning as a team.

Instructors may hold concern that assigning diversified teams could increase group members' perceived conflict. Teams are formed to benefit from diverse expertise and experience, yet conflict states and conflict processes play a large role in determining team effectiveness, particularly in the source and intensity of their perceived incompatibilities (DeChurch et al., 2013). Earlier studies have shown that diversity in teams poses risk of relationship conflict (De Dreu and Weingart, 2003). Other researchers found that neither ethnic diversity nor cognitive diversity influenced task conflict, relationship conflict, team satisfaction, or team performance (Hua, 2013). However, gender diversity was found to significantly coincide with task conflict. Further, task conflict and relationship conflicts were found to be highly detrimental to both team satisfaction, and team performance (Hua, 2013).

Early studies indicated that diversity in groups created relationship conflict due to issues of distrust (Milliken and Martins, 1996). Other studies also indicated that diversity in teams posed the risk of relationship conflict (De Dreu and Weingart, 2003). Researchers investigated further the impacts of specific types of diversity on group conflict. Gender diversity was found to relate positively to group innovation, but only under specific circumstances where the group task had a high degree of novelty (Díaz-García et al., 2013). Hua's (2013) research showed that gender diversity strongly impacted group conflict, while ethnic diversity did not have an influence on group conflict. Gender diversity had a significant positive relationship with task conflict which was found to be highly negatively correlated with both team satisfaction and team performance (Hua, 2013).

Closely related to the theory that diversity increases conflict in teams is the concept that diversity reduces cohesiveness of teams. The impact of diversity on team effectiveness, creativity and innovation, has been inconsistent. Diversity must be carefully managed in order to leverage the potential of teams (Tang, 2019). In order to improve team performance, members must also have familiarity along with interpersonal diversity (Huckman and Staats, 2011).

Studies of team identity showed that groups whose members feel they are equal and connected cooperate better, while those who see their team members as unequal and independent more frequently utilize an avoidance style of conflict management (Boroş et al., 2010). Lower levels of diversity in teams led to more cooperation and less avoidant conflict management strategies as well as less third-party interventions. High levels of group diversity coordinated with a sense of interconnected members led to more cooperative conflict resolution strategies (Boroş et al., 2010).

As the use of group work in college course work increased, so has the frequency of reported social loafing (Hall and Buzwell, 2013). Social loafing is considered free-riding behavior that contributing group members find frustrating when noncontributing members receive the same grade (Hall and Buzwell, 2013). Despite the popularity of teams in college and organizational settings, their productivity has at times been reduced by dishonest actions, including social loafing (Mihelič and Culiberg, 2019). Students who are collaborative learners benefit immensely from group projects, however group work can suffer from social loafing problems often addressed by psychology researchers (Swaray, 2012).

Researchers have identified diffusion of responsibility, attribution of blame, and dehumanization via technology as factors that impact members' tendency to participate in social loafing (Alnuaimi et al., 2010). Additionally, researchers studying the impact of group work within

organizations have found that group members' motivations can impact tendencies for social loafing. When managers adopt exchange theory, people's tendency to respond in reciprocity to support (Eisenberger et al., 1986), as a motivational concept, group members' social loafing can be reduced (Byun et al., 2020).

Researchers have also identified that group conflict can be associated with social loafing (Singh et al., 2018). Contribution conflict, as well as negative emotions of group members, was found to be positively correlated with social loafing. Further research should investigate whether conflict, because of diversity of team members, impacts social loafing.

People working together may function as a group of individuals sharing information or as a team that strives to achieve one common outcome. Several researchers have identified similar factors that differentiate groups and teams. Groups are defined as two or more individuals interacting in their work, while teams are people working together with commitment, shared responsibilities, and complementary skills to achieve shared outcomes and common goals (Hackman, 1990; Robbins and Judge, 2017). Teams also have elements of complementary skills amongst its members (Forsythe, 2014).

The question of whether member diversity impacts functioning as a group or team could be due to member composition. In a team, members are dependent on each other to complete work and share responsibility for outcomes. Even more important to diversity of the members, team members see themselves as an intact social identity embedded into a larger social system (Cohen and Bailey, 1997). Adding instructor-assigned diversity could decrease the members' perception of themselves as having a shared social identity.

Because a core element of functioning as a team involves sharing a common purpose, it could be hypothesized that self-selected student teams may enjoy a greater sense of being a team. Team members are forced to rely on each other's complementary skills, which may be more possible in self-selected rather than instructor-designed teams (van Den Hout et al., 2018). Further, researchers have identified that creative teams must achieve a level of trust and safety to establish a common identity and utilize the knowledge of all members (Paulus and Zee, 2004). Given the short duration of a semester class project, the hypothesis was established that self-selected groups would perceive themselves more as teams than instructor-designed diverse groups.

## Procedure

This study was IRB approved, and undergraduate students from a private university in the Northeastern region of the United States were recruited through convenience sampling. The participants were enrolled in concurrent sections of a management course in two subsequent fall semesters. Surveys were distributed in class. The confidentiality and privacy of the participants were ensured as no identifying information (i.e. name, student I.D. number, etc.) was collected. The survey took the participants about 5–10 minutes to complete.

## Methodology

As this study sought to analyze the impact of a team diversity and formation on members' perceived conflict, cohesiveness, social loafing, and team functioning. The research questions focused on: To what degree does team formation method impact (1) members' conflict; (2) members' cohesiveness; (3) members' social loafing; and (4) members' functioning as a team or group. The study

also investigated whether team diversity and formation had any relationship to members' expected academic grade on the group term project.

The overarching research questions were:

*To what degree does team formation method impact members' perceived group conflict?*

*To what degree does team formation method impact members' perceived group cohesiveness?*

*To what degree does team formation method impact members' perceived social loafing?*

*To what degree does team formation method impact members' perceived functioning as a team or group?*

From the research questions, the following hypotheses were developed:

Hypothesis 1: Instructor-designed diversified teams is positively related to (increases) group members' perceived conflict.

Hypothesis 2: Instructor-designed diversified teams is negatively related to (decreases) group members' perceived cohesiveness.

Hypothesis 3: Instructor-designed diversified teams is negatively related to (decreases) group members' social loafing.

Hypothesis 4: Instructor-designed diversified teams is negatively related to (decreases) members' functioning as a team.

## Method

The questions asked student to use a scale of 1–5, with 1 indicting very little and 5 indicating very much. The survey data responses to each question were analyzed using a t-test of two samples assuming unequal variances to determine whether the means of instructor-designed teams and self-selected teams indicated a statistically significant difference. Students' perceptions of cohesiveness were measured regarding how well the team members got along. The question asked student to use a scale of 1–5, with 1 indicting very little and 5 indicating very much, "How well did group members get along (like each other)?." Students' observations of social loafing were measured in perceptions of group members' loafing. The question asked student to use a scale of 1–5, with 1 indicting very little and 5 indicating very much, "How much social loafing occurred amongst your group members?." The students had awareness of the "social loafing" terminology, as the concept was covered in the course. The survey data responses to the question were analyzed using a t-test of two samples assuming unequal variances to determine whether the means of instructor-designed teams and self-selected teams indicated a statistically significant difference. Students' perceptions of members' functioning as a team were measured in a dichotomous question. The survey question asked student to use a scale of 1–5, with 1 indicting very little and 5 indicating very much, "Did you feel you functioned as a group or a team?." The students had awareness of the difference in definitions of groups and teams, as the theories were covered in the course.

To test the hypotheses, a quantitative approach was taken. To test the positive relationship between a team diversity and formation and student team members' perceived group conflict (Hypothesis 1), student team members' perceived group cohesiveness (Hypothesis 2), student team

members' social loafing (Hypothesis 3), and members' functioning as a team (Hypothesis 4), a quantitative study was conducted. Post-course surveys were completed to measure students' observations of their teams' conflicts, perceptions of team members' cohesiveness, and observations of members' social loafing.

## *Population and sample*

The population for the of the study consisted of students enrolled in concurrent sections of an upper-level organizational behavior course at a small private university in southwestern New York state. Data was gathered from the students in two subsequent fall semesters. All students enrolled in the course were offered the quantitative surveys to complete during class meeting times.

The participants in the study were undergraduate students, in their third or fourth year of college. The course required that students had completed at least 60 hours of academic coursework to enroll in the class. Their majors included business disciplines as well as engineering, fine arts, and liberal arts. A total of 117 students were participants in the study. Group size was three to five students. Instructor-designed diversified groups were estanlished based on diversity of gender, race, and academic major.

## *Instrumentation and data collection procedures*

The instrumentation for the quantitative portion of the study was a paper survey. The invitation to complete the survey was delivered during class time of the last week of the term. The post-course survey instruments contained nine closed-ended scaled questions and one closed-ended multiple choice question. Opening questions on the survey qualified participants for which section of the course they were enrolled and the number of students in their term project team. An additional control question asked the number of team members the respondent considered to be a friend prior to enrolling in the course.

The questions addressing the Hypothesis 1 asked students to rate their level of perceived conflict within their group over work assignments, meeting times, how to complete tasks, making decisions, and personalities. The questions addressing Hypothesis 2 asked students to rate their observations of diversity of views expressed and their perceptions of how well group members got along. The question addressing Hypothesis 3 asked students to rate their observations of how much social loafing occurred amongst group members. Students were familiar with the definition of social loafing as it was a concept taught in the course. The last two questions asked students to indicate whether they functioned as a group or a team (Hypothesis 4) and what grade they expected on the group project.

The five-point scale questions, rating from very little to very much, measured participants' perception of the constructs. The two categorical questions were included at the end of the survey to collect course performance data in order to test for bias. A paper survey distributed in person was determined to be most appropriate to collect responses from first year students.

## *Variables*

The independent variable was defined as the group assignment method—instructor-designed diverse teams. Completion of the course was required since the study evaluated student responses

at the end of the term. Students' success in the course was not a factor of the independent variable, as students completed their surveys, regardless of their grade in the course.

Three dependent variables were included in this study. For Hypothesis 1, the dependent variable was defined as the students' perception of group conflict. For Hypothesis 2, the dependent variable was defined as the students' perception of group cohesiveness. For Hypothesis 3, the dependent variable was defined as the students' observations of social loafing. For Hypothesis 4, the dependent variable was defined as the students' perceptions of their members functioning as a team. Two control variables were established for the study: (1) the size of the group; and (2) the participants' grade expected in the course.

### *Data analysis plan*

The survey data from the quantitative study (Hypothesis 1, 2, 3, and 4) was analyzed using SPSS version 21. Using SPSS's table views, the data was reviewed to confirm that all variables transferred properly with coding in place. For all statistical testing, the level of significance of 0.05 was established. Statistical analyses included descriptive statistics, such as frequencies, mean, median, and standard deviations for all variables contained in the dataset. Observations for the dependent variables were analyzed using responses to questions asking participants to indicate perceptions of conflict and cohesiveness and their observations of social loafing. The control variables were also analyzed to identify any possible correlations with group size or expected grade.

### *Limitations*

The most significant limitation of this study was that participants self-reported their perceptions of conflict, cohesiveness, and functioning and their observations of social loafing. Since the study was conducted within a small university, the findings may not be generalizable to larger populations of courses with group assignments. Additionally, the study was limited to the timeframe in which participants responded to the survey.

The data that support the findings of this study are available from the corresponding author, DM, upon reasonable request. The data collection instrument is publicly available in Open Science Framework (Maguire, 2022).

## Results

The survey data was first analyzed by reviewing the control variables, to ensure that the distribution of respondents was not skewed. An overview of the sample data for size of the group is available in Figure 1. An overview of the sample data for grade expected is available in Figure 2. Secondly, the initial data testing process was to conduct a comparison of means on the variables. The comparison of means included testing (a) between the independent variable and the dependent variables, (b) between the control variables and the dependent variables, and (c) between the control variables and the independent variable. A t-test for comparison of means was utilized.

Results for Hypothesis 1: Instructor-designed diversified teams are positively related to (increases) group members' perceived conflict.

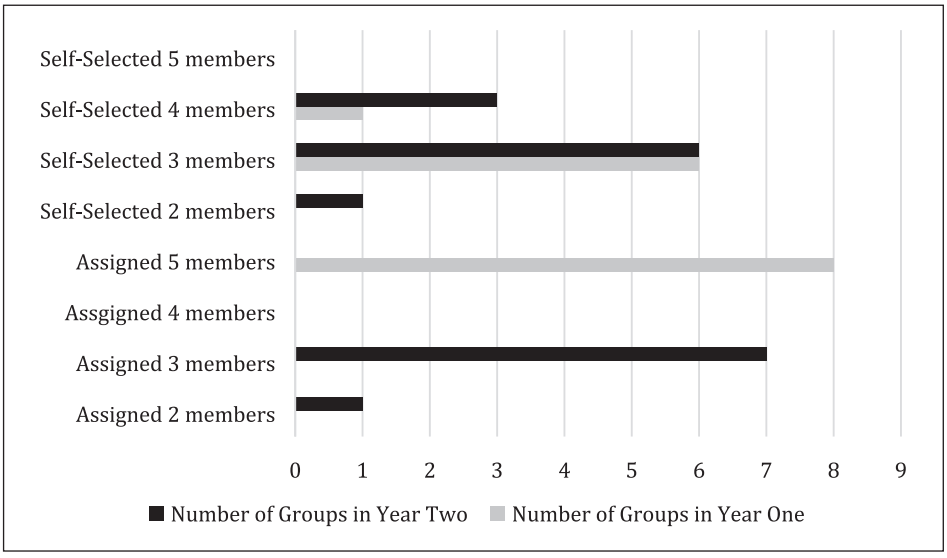



**Figure 1.** Overview of number of members per groups.

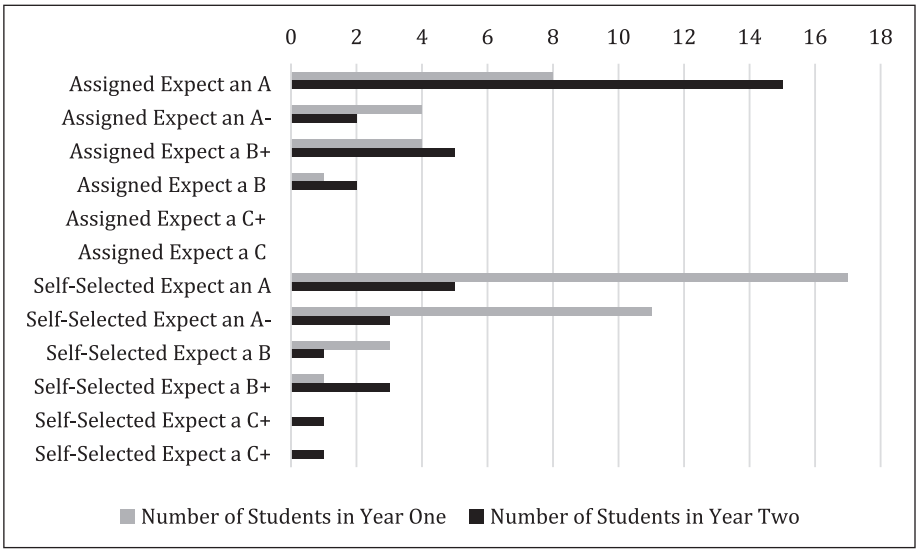



**Figure 2.** Overview of students' grades expected.

Students' perceptions of conflict were measured regarding five types of conflict: (1) conflicts over work assignments, (2) conflicts over meeting times, (3) conflicts over task processes, (4) conflicts over decision-making, and (5) personality conflicts. The research findings reveal no significant difference for conflicts over work assignments, task processes, decision-making, or personality. However, the t-tests did indicate a significant difference for conflict over meeting times. The data indicated that conflict over meeting times was significantly higher in self-selected groups. See Table 1.

**Table 1.** Summary of analysis of team conflict. I-test: Two-sample assuming unequal variances.

| | Assignments | | Meetings | | Complete tasks | | Making decisions | | Personality | |
|---|---|---|---|---|---|---|---|---|---|---|
| | Assigned | Self-select | Assigned | Self-select | Assigned | Self-select | Assigned | Self-select | Assigned | Self-select |
| Mean | 1.394736842 | 1.565217391 | 1.526315789 | 2.282608696 | 1.526316 | 1.76087 | 1.5 | 1.717391 | 1.342105 | 1.456522 |
| Variance | 0.677809388 | 0.740096618 | 0.688477952 | 1.762801932 | 0.796586 | 1.252657 | 0.743243 | 1.14058 | 0.609531 | 0.653623 |
| Observations | 38 | 46 | 38 | 46 | 38 | 46 | 38 | 46 | 38 | 46 |
| Hypothesized mean difference | 0 | | 0 | | 0 | | 0 | | 0 | |
| df | 80 | | 77 | | 82 | | 82 | | 80 | |
| t Stat | -0.925566382 | | 3.183449601 | | -1.06842 | | -1.03223 | | -0.65785 | |
| P($T \leqslant t$) one-tail | 0.178727527 | | 0.001050154 | | 0.144232 | | 0.152501 | | 0.25626 | |
| t critical one-tail | 1.664124579 | | 1.664884537 | | 1.663649 | | 1.663649 | | 1.664125 | |
| P($T \leqslant t$) two-tail | 0.357455055 | | 0.002100308 | | 0.288464 | | 0.305001 | | 0.51252 | |
| t critical two-tail | 1.990063421 | | 1.991254395 | | 1.989319 | | 1.989319 | | 1.990063 | |

The mean difference is significant at the 0.05 level.

In the comparison of means for Hypothesis 1, meeting time conflict, a statistically significant difference was found between the means of students' perceived conflict in meeting times for students in instructor-designed diverse teams and self-selected teams. The mean of assigned students' responses to the question for Hypothesis 1 (meeting times) was 1.52. The mean of self-selected students' responses to the question for Hypothesis 1 (meeting time) was 2.28. With $p < 0.05$, the t-test test rejected the null hypothesis that the two sample sets shared equivalent measures of central tendency. The higher score for students' conflict in meeting times in self-selected teams indicated a significant difference from the score for students' conflict in meeting times in instructor-designed diverse teams reported in survey data. (See Table 1)

Results for Hypothesis 2: Instructor-designed diversified teams are negatively related to (decreases) group members' perceived cohesiveness. Based on t-test variance analysis, the research findings showed no significant difference for cohesiveness. (See Table 2).

**Table 2.** Summary of analysis of team cohesiveness. t-test: Two-sample assuming unequal variances.

| | Assigned | Self-select |
|---|---|---|
| Mean | 4.578947 | 4.347826 |
| Variance | 0.358464 | 0.631884 |
| Observations | 38 | 46 |
| Hypothesized mean difference | 0 | |
| df | 81 | |
| t Stat | 1.518372 | |
| P($T \leqslant t$) one-tail | 0.066407 | |
| t critical one-tail | 1.663884 | |
| P($T \leqslant t$) two-tail | 0.132814 | |
| t critical two-tail | 1.989686 | |

The mean difference is significant at the 0.05 level.

The survey data responses to the question were analyzed using a t-test of two samples assuming unequal variances to determine whether the means of instructor-designed teams and self-selected teams indicated a statistically significant difference. The t-tests showed no significant difference for cohesiveness. (See Table 2)

Results for Hypothesis 3: Instructor-designed diversified teams are negatively related to (decreases) group members' social loafing. The t-test scores disclose a significant difference for social loafing with the observations being higher in the groups that self-selected members. The mean of assigned students' responses to the question for Hypothesis 3 (social loafing) was 1.95. The mean of self-selected students' responses to the question for Hypothesis 3 (social loafing) was 2.644. With $p < 0.05$, the t-test test rejected the null hypothesis that the two sample sets shared equivalent measures of central tendency. The higher score for students' social loafing in self-selected teams indicated a significant difference from the score for students' social loafing in instructor-designed diverse teams reported in survey data. (See Table 3).

**Table 3.** Summary of analysis of social loafing. t-test: Two-sample assuming unequal variances.

| | Assigned | Self-select |
|---|---|---|
| Mean | 1.947368421 | 2.644444444 |
| Variance | 1.132290185 | 1.325252525 |
| Observations | 38 | 45 |
| Hypothesized mean difference | 0 | |
| df | 80 | |
| t Stat | 2.863824192 | |
| P($T \leqslant t$) one-tail | 0.002671495 | |
| t critical one-tail | 1.664124579 | |
| P($T \leqslant t$) two-tail | 0.005342991 | |
| t critical two-tail | 1.990063421 | |

The t-tests showed significant difference for social loafing with the observations being higher in the groups that self-selected members. The mean of assigned students' responses to the question for Hypothesis 3 (social loafing) was 1.95. The mean of self-selected students' responses to the question for Hypothesis 3 (social loafing) was 2.644. With $p < 0.05$, the t-test test rejected the null hypothesis that the two sample sets shared equivalent measures of central tendency. The higher score for students' social loafing in self-selected teams indicated a significant difference from the score for students' social loafing in instructor-designed diverse teams reported in survey data. (See Table 3)

Results for Hypothesis 4: Instructor-designed diversified teams are negatively related to (decreases) members' functioning as a team. . The research findings using a Z-test comparing two proportions showed significant difference for groups verses teams with the proportion being higher for functioning as a team in the groups that had instructor-assigned members and the proportion being higher for functioning as a group in those that self-selected members. The percentage of assigned students' responses to the question for Hypothesis 4 (group verses team) indicating functioning as a team was 78.94% and the percentage indicating functioning as a group was 21.05%. The percentage of self-selected students' responses to the question for Hypothesis 4 (group verses team) indicating functioning as a team was 39.47% and the percentage indicating functioning as a group was 78.94%. With $p < 0.05$, the Z-test test rejected the null hypothesis that the two sample sets shared equivalent measures of central tendency. The higher score for students' group functioning in self-selected groups and higher score for students' team functioning in instructor-assigned diverse groups indicated a significant difference in the reported survey data. (See Table 4)

**Table 4.** Summary of analysis of perception of team or group. Perception of team or group. Two-tailed Z test comparing two proportions. Two-tailed Z test.

| | Assigned | Self-select |
|---|---|---|
| Percentage team | 0.789473684 | 0.394736842 |
| Percentage group | 0.210526316 | 0.789473684 |
| z-value= | 3.624931108 | |
| *p*-value | 0.000289038 | |

The survey data responses to the question were analyzed using a Z-test comparing two proportions to determine whether the percentage of instructor-designed teams and self-selected teams indicated a statistically significant difference. The Z-test showed significant difference for groups verses teams with the proportion being higher for functioning as a team in the groups that had instructor-assigned members and the proportion being higher for functioning as a group in those that self-selected members. The percentage of assigned students' responses to the question for Hypothesis 4 (group verses team) indicating functioning as a team was 78.94% and the percentage indicating functioning as a group was 21.05%. The percentage of self-selected students' responses to the question for Hypothesis 4 (group verses team) indicating functioning as a team was 39.47% and the percentage indicating functioning as a group was 78.94%. With $p < 0.05$, the Z-test test rejected the null hypothesis that the two sample sets shared equivalent measures of central tendency. The higher score for students' group functioning in self-selected groups and higher score for students' team functioning in instructor-assigned diverse groups indicated a significant difference in the reported survey data. (See Table 4)

## Discussion

According to the findings of this study, team formation and diversity had a significant impact on teams. The analysis of team formation and diversity on team conflict indicated significant impact regarding social loafing and functioning as a team. Instructor-designed diverse teams did not impact perceived team conflict or cohesiveness. Self-selected student teams perceived no more cohesiveness than instructor-designed diverse teams.

The association of diversity with team conflict was not substantiated in this study. This finding disputes the suggestions by earlier reseachers whose studies indicated that diversity in teams posed the risk of relationship conflict (De Dreu and Weingart, 2003; Milliken and Martins, 1996). Students' perception of team conflict was not higher in the instructor-designed diverse groups. The findings in this study counter the previous findings that minimizing diversity could increase cohesiveness or reduce conflict (O'Reilly et al., 1989; Smith et al., 1994; Wiersema and Bantel, 1992)The analysis showed no difference between self-selected teams and instructor-designed diverse teams for perceptions of conflict over assignments, processes, decision-making or personalities. However, the analysis showed that conflict over meeting times was significantly higher amongst self-selected groups. Since instructor-designed teams were established to increase diversity of majors, genders, and races, the potential for conflict in available meeting times would have been understandable. However, since the conflict in meeting times was higher in self-selected groups, students may have taken advantage of their familiarities when committing time to their group project.

The analysis of cohesiveness amongst self-selected and instructor-designed diverse student groups also yielded somewhat surprising results. One might expect that students who selected their group members would indicate a higher level of cohesiveness, as in a stronger indication that group members liked each other. However, the data showed no significant difference between self-selected and instructor-designed diverse groups. It may be possible that because neither set of groups perceived personality conflicts, as discussed in hypothesis one, their acceptance of members was not impacted. It is also possible that the students in the study were open to accepting work with diverse students as well as work with classmates with whom they had a preestablished relationship, or prior knowledge. The findings may reinforce previous research that suggested diversity, when coupled with familiarity, improves team performance (Huckman and Staats, 2011).

The results indicated that social loafing by group members varied between the self-selected and instructor-designed diverse groups. The data indicated that social loafing was higher amongst self-selected groups than in instructor-designed diverse groups. This finding could indicate that

students in self-selected groups were more familiar with each other, and thus, were more likely to take advantage of the preestablished relationship by slacking off on contributions to the group work. However, the level of perceived cohesiveness did not differ between the two groups. Singh et al. (2018) identified that group conflict can be associated with increased social loafing. However, this study found only an increase in meeting-time conflict and an increase in social loafing in self-selected teams. The findings could suggest that students in instructor-designed diverse groups felt a greater sense of need to contribute to the group work. It is possible that the instructor-designed groups felt their role in the group had greater identity or significance since they were assigned to a group with greater diversity of student backgrounds.

The study's findings indicated that students' perceptions of their functioning as a team, with shared goals and accountability, was significantly higher amongst students in instructor-designed diverse groups than in self-selected groups. This finding reinforces earlier findings that diversity increases team performance when members have elements of complementary skills (Forsythe, 2014). Students in self-selected groups indicated to a much higher degree that they functioned as a group, working alongside each other, and sharing information, but lacking in synergy or collective accountability. These results were somewhat surprising in that the self-selected student groups could have had a greater sense of obligation to work together as a team due to their preestablished familiar relationships. However, the instructor-designed diverse teams may have had the advantage of perceived uniqueness of each member and their contributions. The instructor-designed diverse teams may have felt a greater dependency on each other to contribute toward their project goals.

The findings in this study suggested that instructor-designed diverse teams did not increase member conflict, while student-selected teams did not improve team cohesiveness. Instructor-designed diverse teams did impact social loafing and the members' functioning as a team. This study's findings that instructor-designed diverse teams had advantages of decreased conflict, decreased social loafing, and increased perception of team functioning offers significant value to instructors utilizing team assignments in coursework. Students have the tendency to want to work with their friends on group assignments. However, this study indicated that instructor-designed diverse groups and offer advantages. The extra level of effort for the instructor to facilitate the assigning of diverse groups is outweighed by the potential to decrease conflict and social loafing, while increasing team concept perceptions.

This study provides a better understanding of team formation and diversity on team conflict. The study offers insight on the advantages of instructor-designed teams that encourage diversity of students' academic disciplines and demographics. Instructors seeking to improve student group work experience should consider the findings of this study useful in their group project structures. This study also indicates that students should be made aware of the limitations of self-selected teams for group projects. Students should be encouraged to accept instructor-designed diverse teams to improve their group work experiences.

Going forward, researchers should continually monitor and assess the formation of student teams, not only in management courses, but all fields. Researchers should include a variety of group project types, as well as the use of technology amongst the student groups.

## Declaration of conflicting interests

The author(s) declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

## Funding

The author(s) received no financial support for the research, authorship, and/or publication of this article.

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